Speech to Indian Sign Language

A Retrieval-Based Speech-to-ISL System

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1. Problem and Motivation

The problem

A person who does not know Indian Sign Language cannot easily communicate with a Deaf person who primarily uses ISL.

Our idea

- Take spoken English as input.
- Convert speech into text.
- Find the corresponding ISL sign.
- Show the sign as real human video.

Example

Speech: "Hello, thank you."

↓

Text: `hello thank you`



2. Project Aim

Main aim

- Build a working speech-to-ISL prototype.
- Use real ISL videos performed by human signers.
- Keep the system simple enough to run on a normal laptop.
- Measure how much of the spoken input can actually be converted.
- Measure the delay between speech and sign output.

Important scope decision

We are currently building a **retrieval system**, not a full ISL translation or sign-generation system.

Phase 0 focus:

Speech Recognition → Text Processing → ISL Video Retrieval

3. Why Not Directly Generate Sign Language?

Indian Sign Language is a complete natural language.
It is not simply English words performed using hand movements.

- ISL has its own grammar and word order.
- Facial expressions and body movements also carry meaning.
- Continuous sign language is difficult to generate naturally.
- Good ISL datasets are still limited compared with speech datasets.

Our design choice

Instead of generating artificial signing, we retrieve **real recorded ISL videos**.

Advantage:
The output is actual human signing rather than an artificial avatar or generated motion.

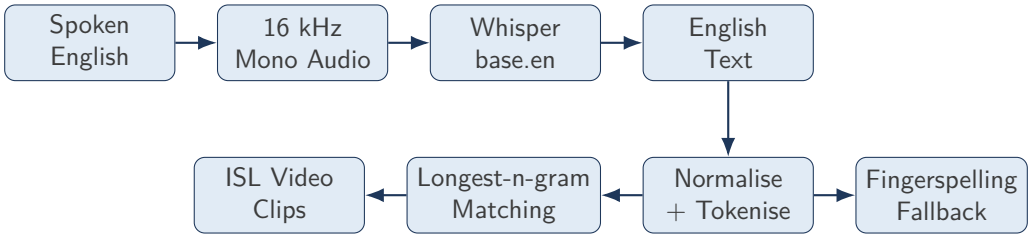
4. Research Background and Gap

Area	Existing work	Observation
Speech Recognition	Whisper	Strong speech-to-text models already exist.
ISL Datasets	ISLTranslate, iSign, INCLUDE	ISL resources are growing, but still limited compared with major languages.
Sign Recognition	Neural Sign Language Translation	Sign $\rightarrow$ text has received considerable research attention.
Sign Generation	Progressive Transformers, speech-to-sign work	Generating natural signing is still difficult.
Retrieval	Sentence-BERT	Semantic retrieval can help in a future phase.

Our gap

We investigate a simpler approach: **speech $\rightarrow$ text $\rightarrow$ retrieval of real ISL videos**.

5. Overall System



Recognise → Clean → Match → Play

6. Dataset and Sign Vocabulary

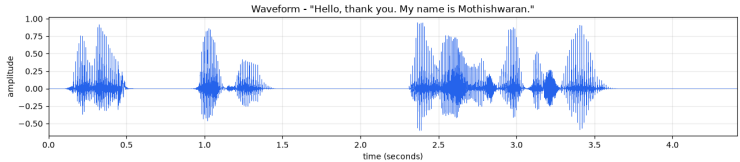
Main source: ISLRTC Indian Sign Language Dictionary

- Real ISL videos from the official ISL dictionary.
- Videos are performed by Deaf signers.
- The complete dictionary contains around 10,000 terms.
- Our current prototype intentionally uses only a very small subset.

Current Phase 0 vocabulary:

Entry	Type
hello	single word
thank you	2-word phrase
sorry	single word
please	single word
name	single word
so far so good	4-word phrase

7. Speech Processing: From Sound to Data



Our audio representation

- Sampling rate: **16 kHz**
- Channels: **Mono**
- Sample representation: **PCM 16-bit**

Why 16 kHz?

By the Nyquist theorem, 16 kHz sampling represents frequencies up to approximately 8 kHz, which is suitable for speech and matches Whisper's input requirement.

8. Framing, Windowing and STFT

Speech changes continuously, so we cannot treat a long recording as one stationary signal.



- Frame length: **25 ms**
- Hop size: **10 ms**
- Overlap: approximately **60%**
- Hamming window reduces spectral leakage.
- FFT moves the signal from the time domain to the frequency domain.

9. Spectrogram and Log-Mel Representation

- **Linear spectrogram:** shows how frequency energy changes over time.
- **Log-Mel spectrogram:** compresses the frequency information into the Mel scale.
- **Whisper uses 80-bin log-mel features** as its input representation.

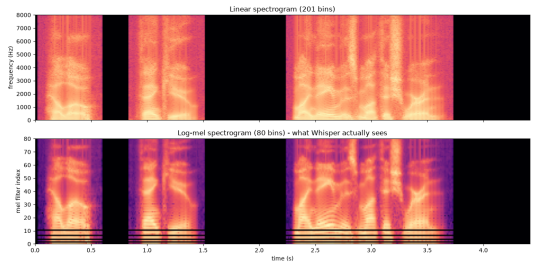


Figure: Enter Caption

10. MFCC vs Log-Mel: An Important Speech Concept

Classical speech-processing pipeline:

Audio → Framing → Windowing → FFT → Mel Filterbank → Log → DCT → MFCC

Why do we discuss MFCCs if Whisper does not use them?

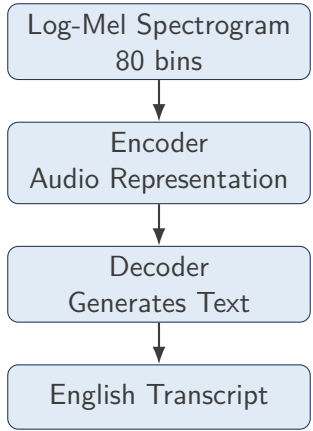
- MFCCs are a fundamental classical speech feature.
- DCT compresses the log-mel features into a smaller representation.
- Classical systems often used MFCCs with GMM-HMM models.
- Whisper is a neural network, so it directly uses the **log-mel spectrogram**.
- Therefore, our implementation studies MFCCs conceptually but does not feed MFCCs to Whisper.

Key point

Our actual ASR input is **80-bin log-mel spectrogram**, not MFCC.

11. Speech Recognition using Whisper

Whisper is an encoder-decoder Transformer for speech recognition.



Our configuration

Setting	Value	Reason

12. Text Normalisation and Tokenisation

Whisper output contains punctuation and capitalisation.
Example:

```
Hello, thank you.  My name is Mothishwaran.
```

We create three forms of the input:

Representation	Example	Purpose
Original	Mothishwaran	Name detection
Normalised	mothishwaran	Dictionary lookup
Sentence boundary	sentence-initial / not	Avoid false name detection

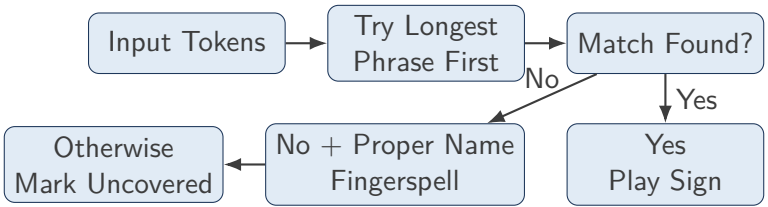
Important implementation case
"My" is capitalised because it starts a new sentence.
Therefore:

$$\texttt{My} \neq \text{proper name}$$

But:

13. Longest-n-gram Matching

The matcher scans the sentence from left to right.



Why longest-first?

- "thank you" should be treated as one phrase.
- "so far so good" is a 4-word phrase and must produce one clip.
- The maximum phrase length is read from the manifest, not hardcoded.

14. Fingerspelling Fallback

If a proper name is not present in the sign dictionary, the system fingerspells it.

Mothishwaran

↓

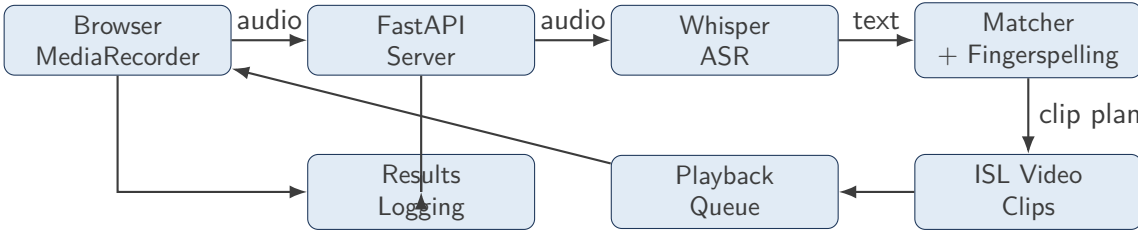
M – O – T – H – I – S – H – W – A – R – A – N

- The system has alphabet clips from A to Z.
- Each letter is played sequentially.
- The ISL alphabet used here is two-handed.
- Non-alphabetic characters are ignored.

Why not fingerspell every unknown word?

Fingerspelling is mainly useful for names and special terms. Fingerspelling every missing word would make the output difficult to read and would hide the actual vocabulary coverage.

15. Complete Implementation Architecture



Implementation technologies

- Python
- Faster-Whisper
- FastAPI
- HTML + JavaScript MediaRecorder
- CSV manifest for the sign vocabulary
- MP4 video clips for ISL output

16. End-to-End Demonstration

Input

"Hello, thank you.
My name is Mothishwaran."

Whisper output

```
hello thank you my name is
mothishwaran
```

Playback plan

- hello → ISL clip
- thank you → ISL clip
- my → uncovered
- name → ISL clip
- is → uncovered
- Mothishwaran → fingerspell

Image placeholder
demo.png
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17. Evaluation: What Do We Measure?

1. Coverage

$$\text{Coverage} = \frac{\text{tokens matched to an ISL sign}}{\text{total tokens}} \times 100$$

It tells us how much of the spoken input exists in our current sign inventory.

2. Word Error Rate (WER)

$$WER = \frac{S + D + I}{N}$$

- Substitution
- Deletion
- Insertion

WER measures the **speech recognition stage**, not the quality of the ISL output.

3. End-to-end latency

Recording stops $\rightarrow$ **first ISL video frame appears**

18. Current Results

Measurement	Current Result
ASR median latency	**833 ms**
ASR range	810–849 ms
Matching latency	**0 ms**
Server total	834 ms
End-to-end latency	**1220 ms median**
End-to-end range	1019–1790 ms
Mean coverage	**55.4%**

Current observations

- Matching is extremely fast compared with speech recognition.
- The main latency bottleneck is the ASR stage.
- The current vocabulary is intentionally very small.
- The system successfully demonstrates the complete pipeline.

19. What We Learned from the First Implementation

Finding 1 – ASR is the main bottleneck

- Matching takes approximately 0–1 ms.
- Speech recognition takes hundreds of milliseconds.
- Therefore, improving the matcher will not significantly improve total latency.

Finding 2 – Proper names are difficult

- Whisper repeatedly produced “Mothish” instead of the full name in some runs.
- The matcher itself correctly processes whatever text it receives.
- This shows that the error can originate before the retrieval stage.

Finding 3 – Fallback paths matter

- Browser-side audio conversion did not always follow the intended path.
- Server-side resampling still allowed the pipeline to work.
- Logging helped us identify this instead of assuming the pipeline worked as designed.



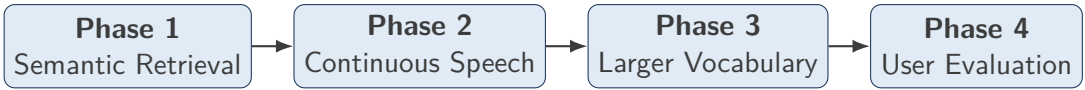
20. Current Limitations

- Current vocabulary contains only six sign entries.
- Exact matching cannot understand paraphrases.
- English word order is retained; this is not full ISL grammar.
- Facial expressions and other non-manual markers are not generated.
- Separate clips do not naturally model co-articulation between signs.
- Unknown words may be left uncovered.
- No evaluation with Deaf signers has been conducted yet.
- Whisper is used as an existing component; our contribution is the retrieval pipeline.

Most important limitation

The current prototype should be described as **speech-driven ISL video retrieval**, not as complete automatic ISL translation.

21. Planned Next Phase



- Use sentence embeddings for meaning-based retrieval.
- Add continuous speech using voice activity detection.
- Expand the sign vocabulary substantially.
- Handle word-sense ambiguity and context.
- Evaluate the system with Deaf ISL users.
- Improve transitions between consecutive signs.

22. Conclusion

What we have achieved

A complete working prototype that takes spoken English and produces a sequence of real ISL signing videos.

Pipeline

Speech → Whisper → Text → Normalisation → Phrase Matching → ISL Video

Current evidence

- End-to-end pipeline works on a laptop CPU.
- Matching is effectively instantaneous.
- Current mean coverage is approximately 55%.
- The main performance bottleneck is speech recognition.

The next goal is not just more output — it is more meaningful and correct ISL output.

References

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Data sources

- Indian Sign Language Research and Training Centre (ISLRTC)
- ISL Dictionary
- FDMSE / RKMVERI Indian Sign Language alphabet resources

Thank You

Questions?

Speech Recognition + Speech Processing + Retrieval + Real ISL Video

